RETAIL-FLOW // Spatial and Technical Infrastructure

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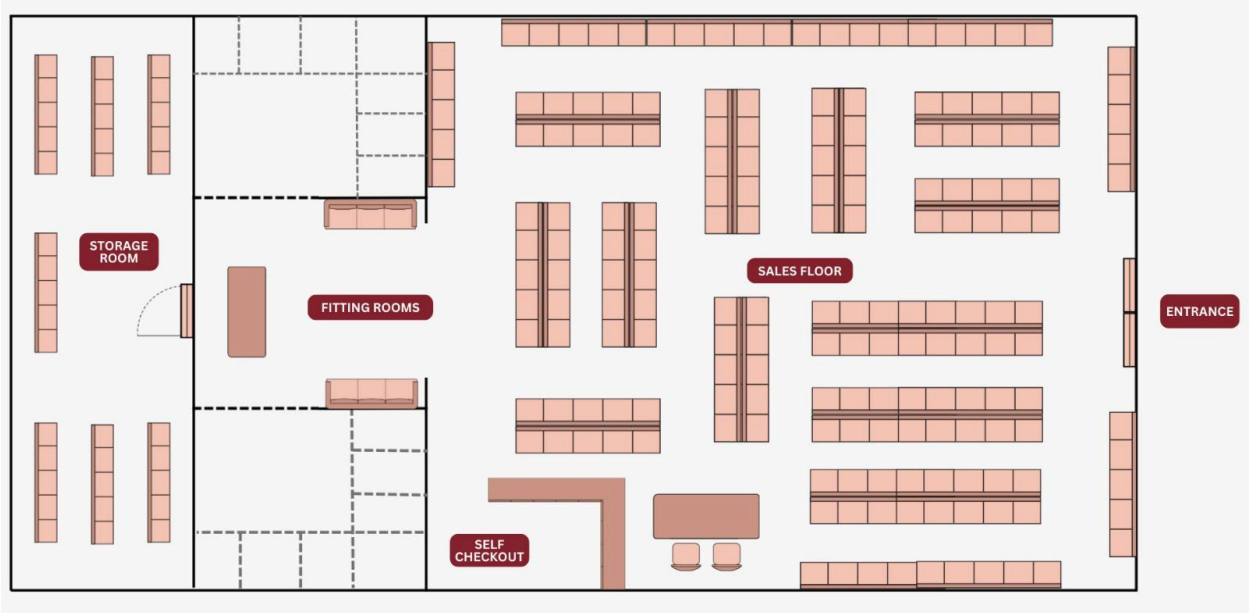
1. Spatial Infrastructure

The store operates within a defined 100 ft x 50 ft surface area that is divided into five key functional zones. This layout is designed to facilitate a natural flow, guiding customers through high engagement areas before reaching the final transaction point.

- **Entrance:** A singular, controlled entry and exit point that will be used to count people going in and out.
- **Sales floor:** This is where we can find the shelving units with the products, making it our primary data source.
- **Fitting rooms:** A small lobby consisting of a customer lounge with sofas and fitting rooms that will help us analyze the products that are capturing the customers' attention.
- **Self-checkout:** This is the final point of the customer where we can capture data focused on the transactions and see what's selling more.
- **Storage:** A restricted area for inventory management.



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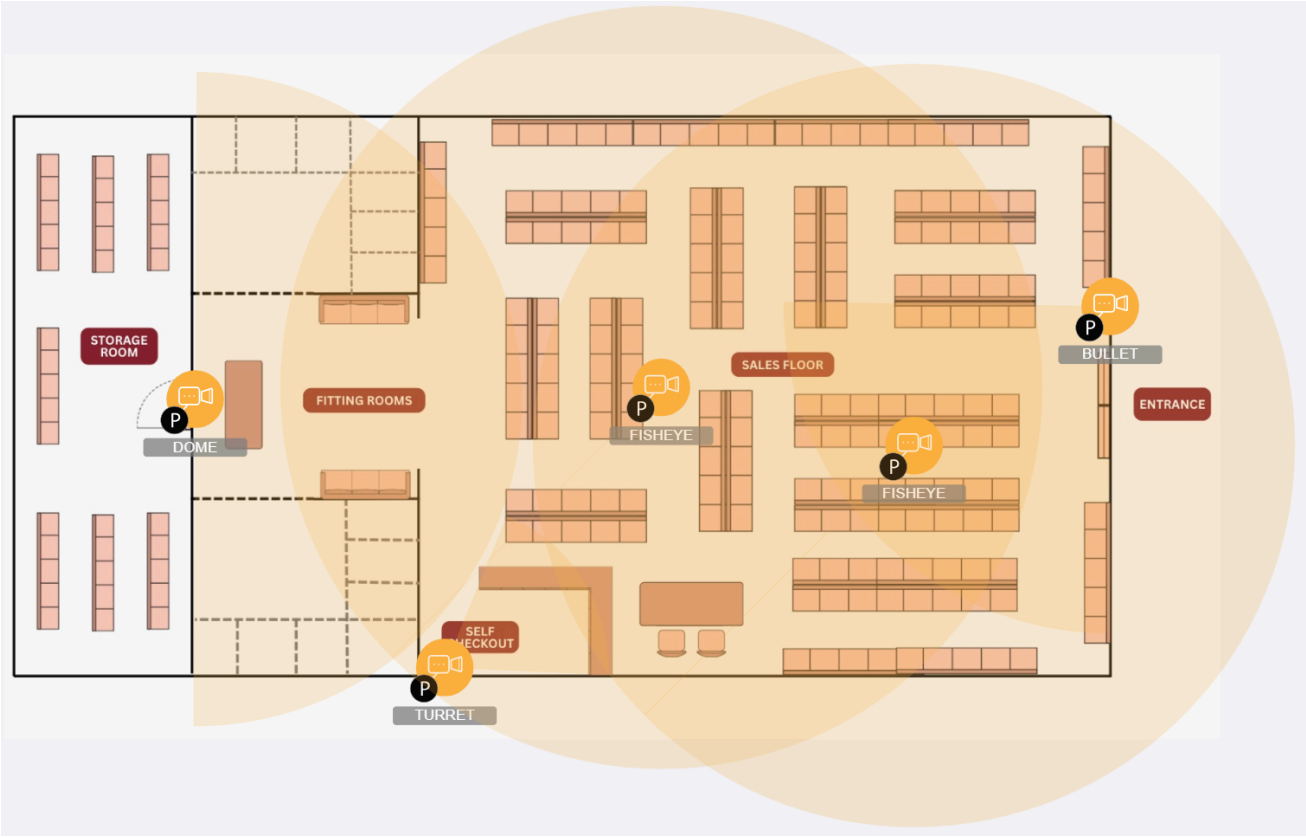
2. Technical Infrastructure

Our system employs an advanced array of five specialized cameras to convert physical movement into a digital data stream. By implementing targeted lens technologies in critical areas, we guarantee precise capture for both security and behavioral analysis.



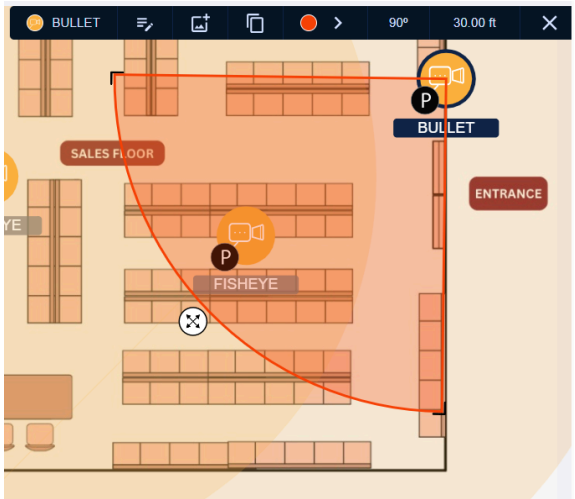
- **Camera 1: Entry/Exit Control**

  Camera Type: Fixed-Lens Bullet

  Viewing range: 90º angle & 30 ft radius

  Placement: Mounted above the primary entrance, oriented toward the interior.

  Purpose: Records entry/exit events to calculate store-wide conversion rates.



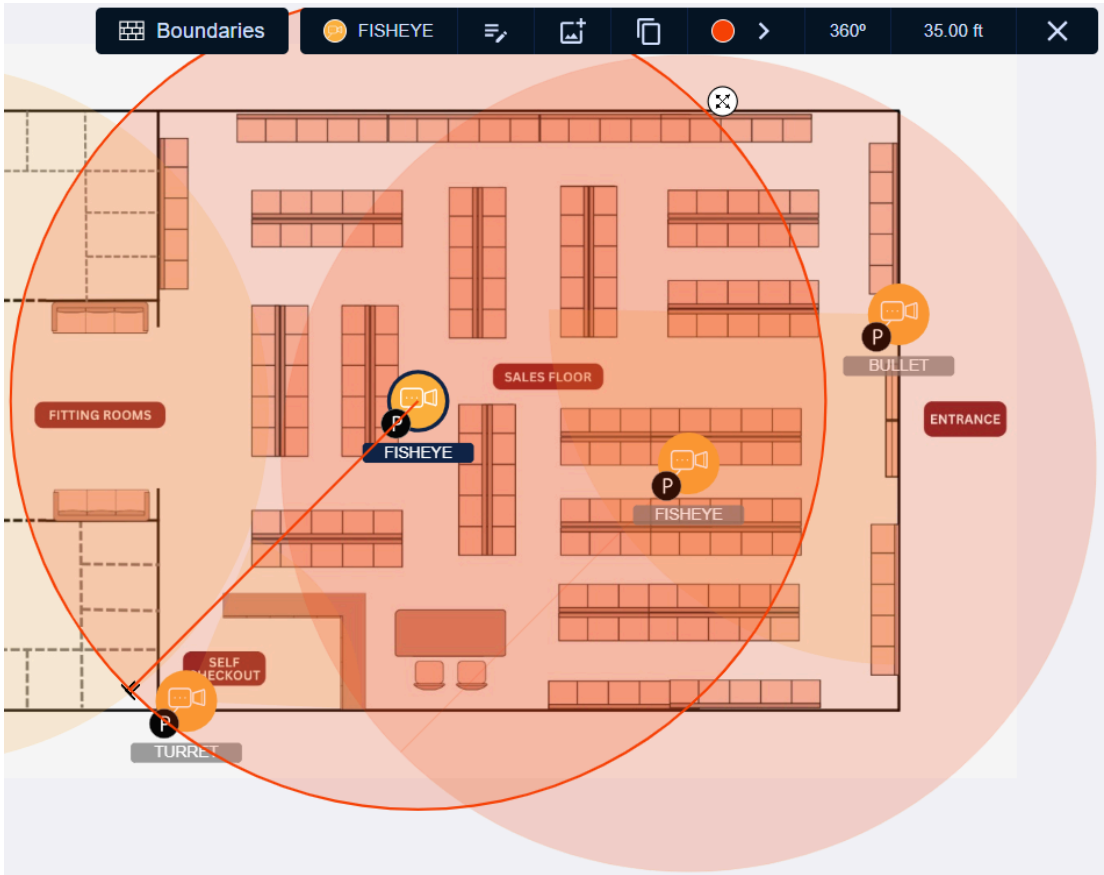
- **Camera 2 & 3: Behavioral Core**

  Camera Type: Multi-Lens Fisheye

  Viewing range: 360º angle & 35 ft radius

  Placement: Mounted in the main sales floor area with an overlap to eliminate shelf shadows.

  Purpose: Facilitates seamless path tracking and heat mapping as customers navigate the store and identify popular and ignored merchandise.



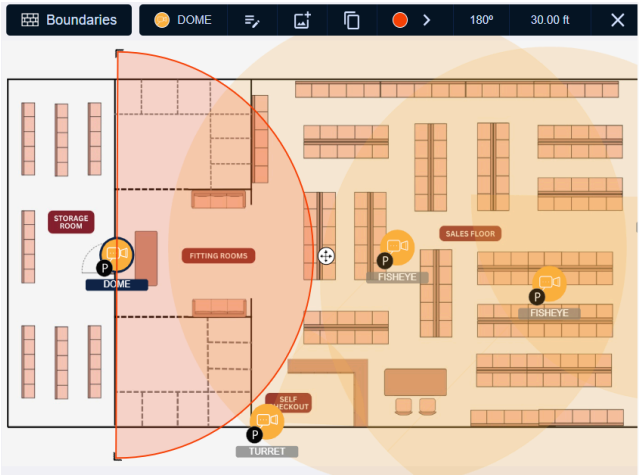
- **Camera 4: Fitting Room Monitor**

  Camera Type: Fixed Dome

  Viewing range: 90º angle & 30 ft radius

  Placement: Mounted in fitting room lobby.

  Purpose: Measures dwell time in the lounge and possible transitions from browsing to trying on.



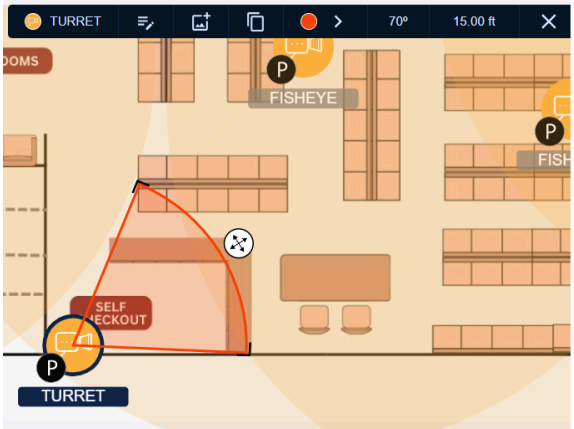
- **Camera 5: Transaction Monitor**

  Camera Type: Fixed-Lens Turret

  Viewing range: 70º angle & 15 ft radius

  Placement: Mounted directly over the self-checkout area.

  Purpose: Correlates physical customer behavior with digital transaction logs to measure the "Popularity vs. Purchase" index.



Works Cited

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